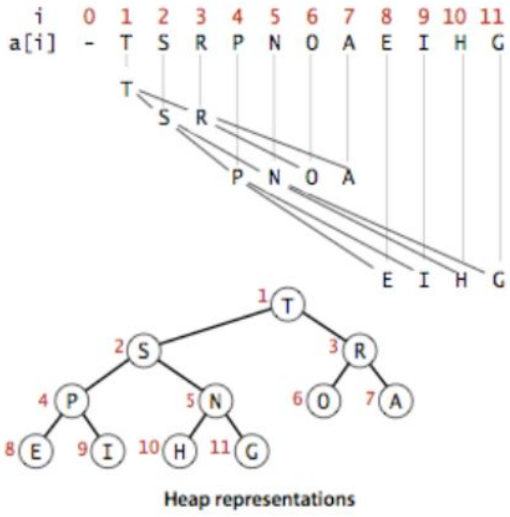
Heapsort - Detyra të zgjidhura

Lënda: Hyrje në Algoritme

Binary Heap:

- Është Data Structure
- Implementon Priority Queues
- Përdoret për sortim (tek heapsort)
- Cdo node mund të ketë maksimalisht **2 fëmijë**
- Druri duhet të jetë **complete binary tree**



Heap representations

HeapSort :

I: Heap Construction - > vendosja e elementeve të vargut në heap

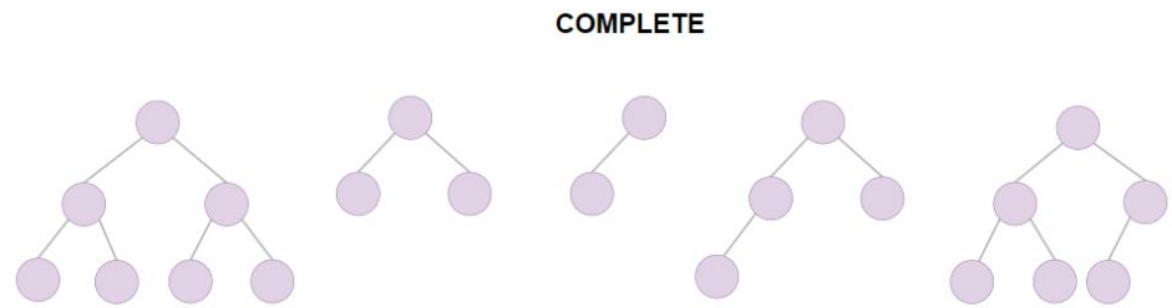
- > Heap duhet të bëhet ordered (e renditur)

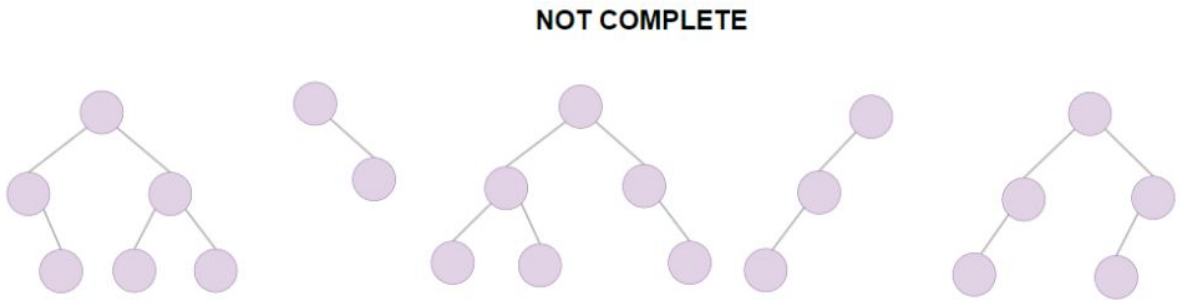
II : Sortdown - > repeatedly delete the largest remaining item

- > pas çdo fshirje, heap duhet të jetë ordered

Complete binary tree:

Është dru i kompletuar (plotësuar), përpos në nivelin e fundit. Në nivelin e fundit, nyjet janë vendosur nga majtas në të djathtë, pa lëshuar hapësira të zbrazëta midis tyre.





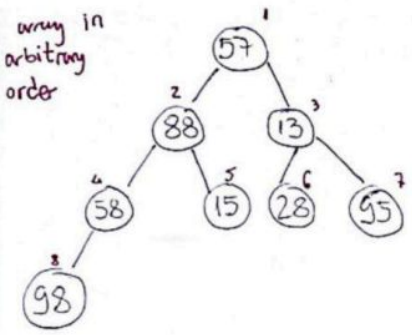
Detyra 1: Të sortohet vargu i dhënë me heapsort:

$a = \{57, 88, 13, 58, 15, 28, 95, 98\}$

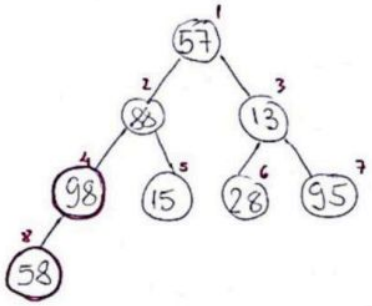
• Fëmijët e nyjës k: $2k$, $2k+1$

. Indeksim fillon prej 1

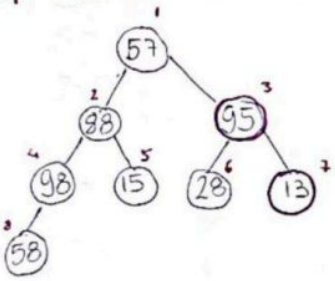
I: Heap Construction



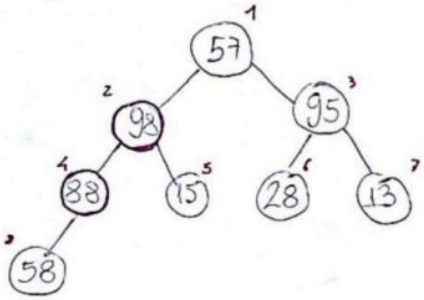
Hapi 1) sink(4); // 58



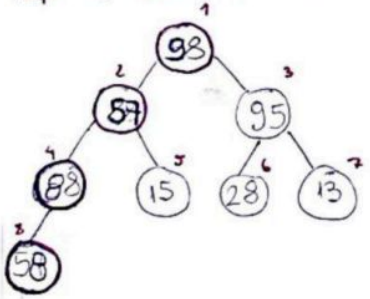
Hapi 2) sink(3); // 13



Hapi 3) sink(2); // 88

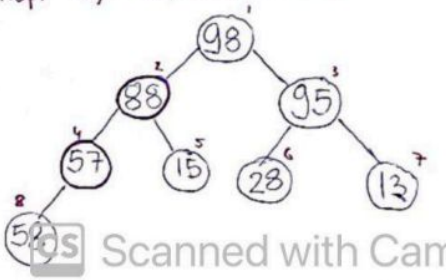


Hapi 4) sink(1) // 57

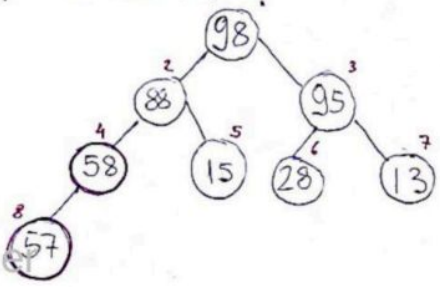


Ordered heap //

Hapi 5) sink(2) // 57

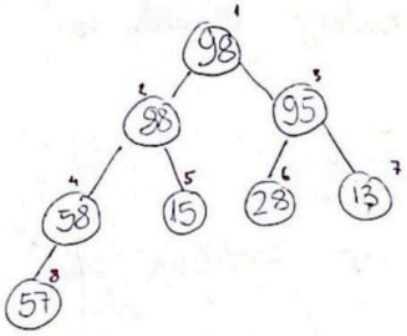


Hapi 6) sink(4) // 57

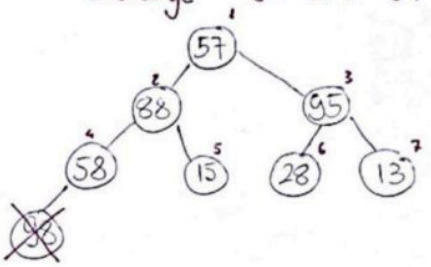


Scanned with CamScanner

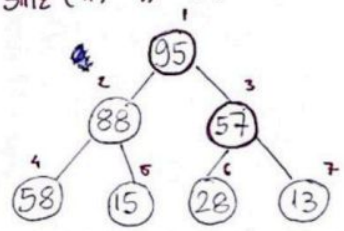
II: Sortdown



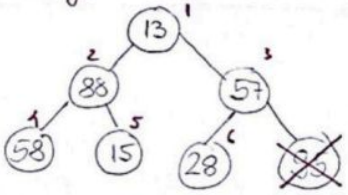
1) delMax(); // 98
exchange 98 and 57



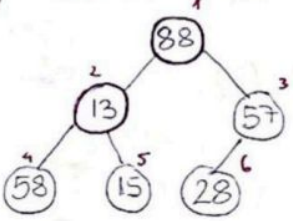
2) sink(1) // 57



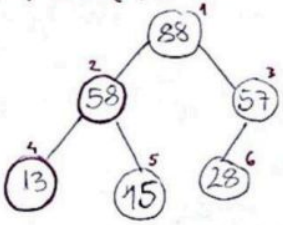
3) delMax(); // 95
exchange 95 and 13



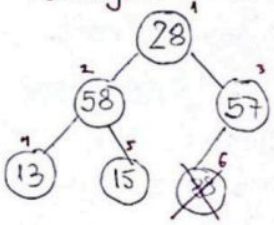
4) sink(1) // 13



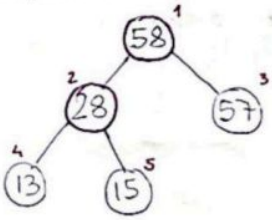
5) sink(2) // 13



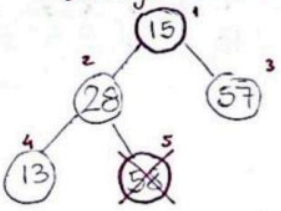
6) delMax(); // 88
exchange 88 and 28



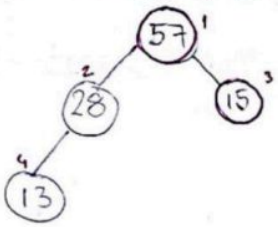
7) sink(1) // 28



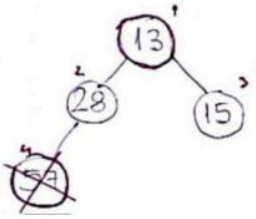
8) delMax(); // 58
exchange 58 and 15



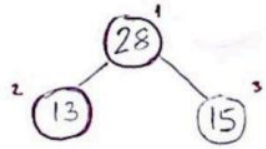
9) sink(1) // 15



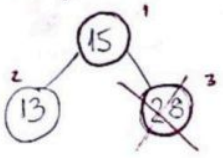
10) delMax(); // 57
exchange 57 and 13



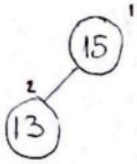
11) sink(1) // 13



12) delMax(); // 28
exchange 28 and 15

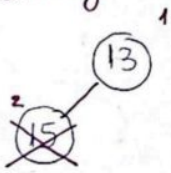


13) sink(1); // it doesn't sink



14) delMax(); // 15

exchange 15 with 13



end of sortdown phase

Array in sorted order:

13,	15,	28,	57,	58,	88,	95,	98

Detyra 2: Nga vargu i dhënë $A = \{13, 28, 21, 9, 4, 78, 49, 68, 19, 5\}$

a) Të krijohet Max-Binary Heap.

b) Nga binary heap i krijuar në hapin e parë (a) të fshihet elementi root.

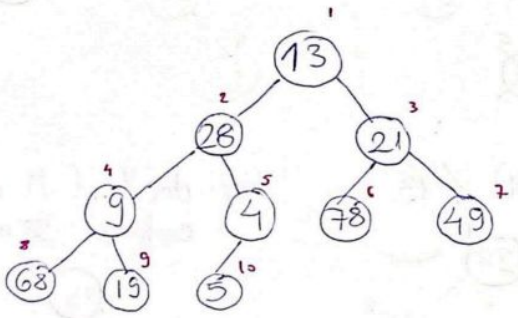
c) Nga binary heap i krijuar në hapin e dytë (b) të shtohet elementi me vlerë 41.

d) Nga MaxBinary Heap i krijuar, të sortohet vargu me HeapSort.

Zgjidhje:

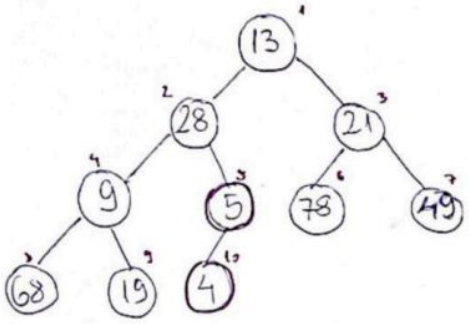
a) Heap Construction

1	2	3	4	5	6	7	8	9	10
13,	28,	21,	9,	4,	78,	49,	68,	19,	5

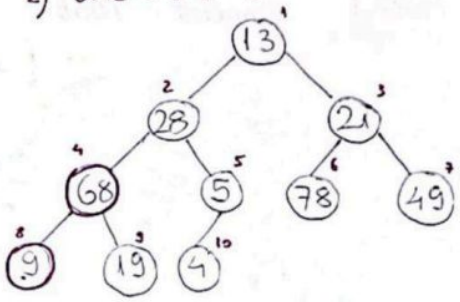


← array in arbitrary order

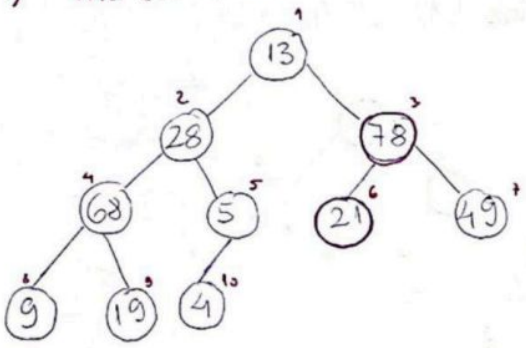
1) sink(5); // 4



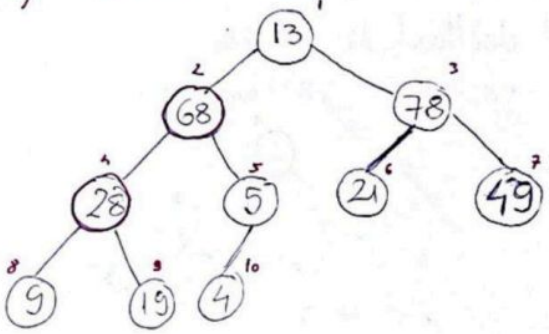
2) sink(4); // 9



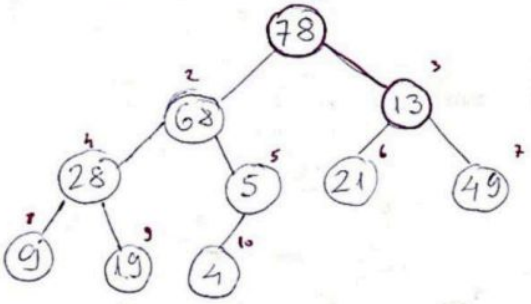
3) sink(3) // 21



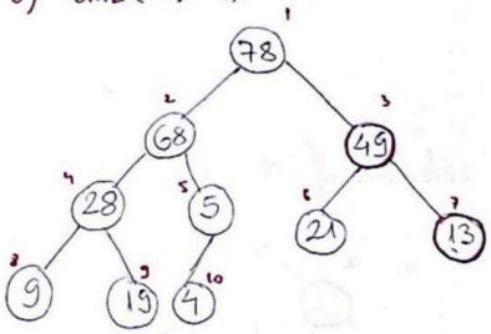
4) sink(2) // 28



5) sink(1) // 13

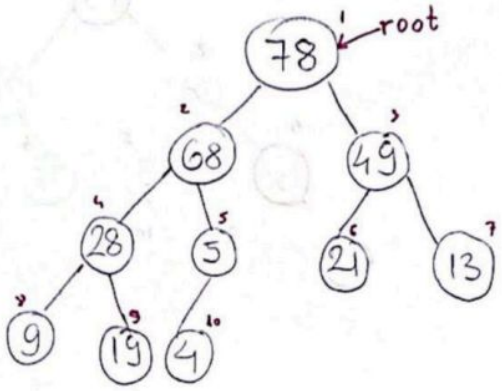


6) sink(3) // 13



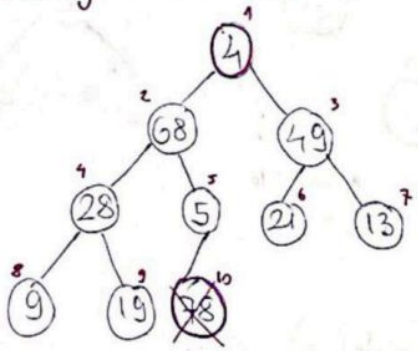
Ordered heap

b) Fshihet elementi root

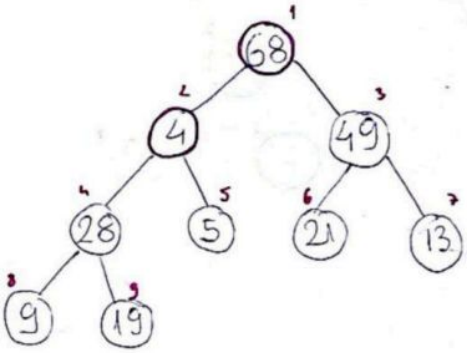


1) delMax(); // 78

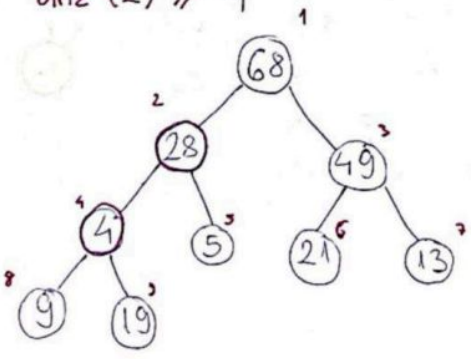
exchange 78 and 4



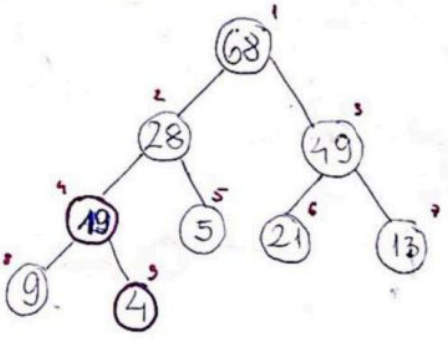
2) sink (1) // 4



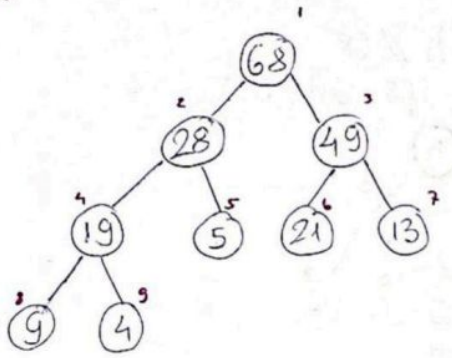
3) sink (2) // 4



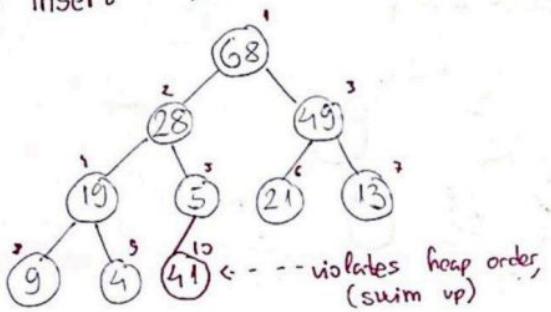
4) sink (4) // 4



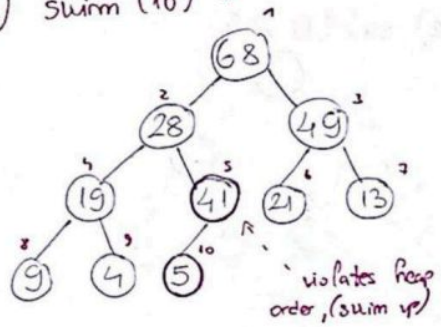
c) Të shtohet elementi me vlerë 41 :



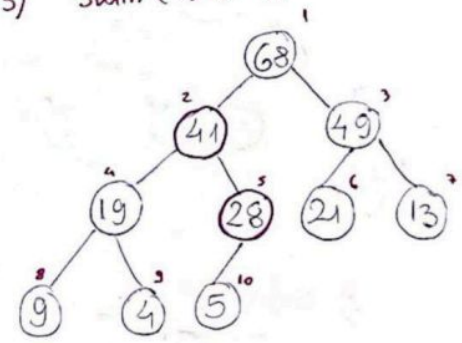
1) insert 41 :



2) swim (10) // 5

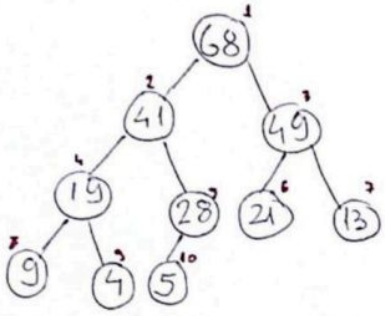


3) swim (5) // 41

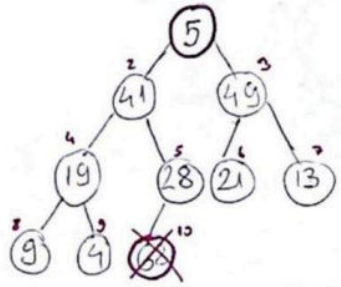


heap ordered //

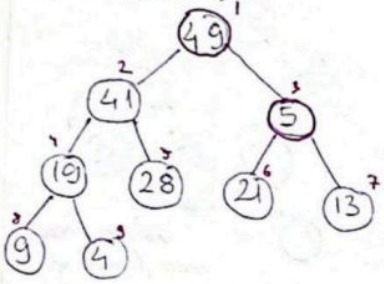
d) Të sortohet vargu me HeapSort



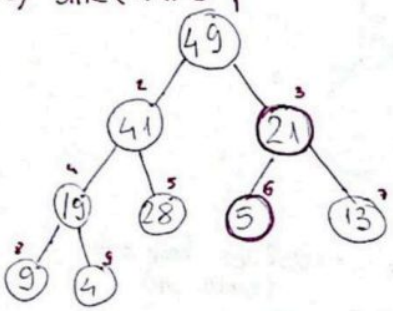
1) delMax(); // 68
exchange 68 and 5



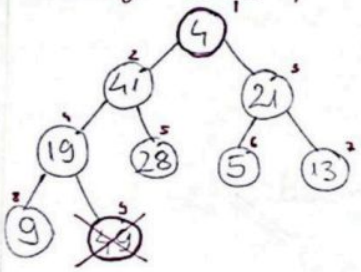
2) sink(1) // 5



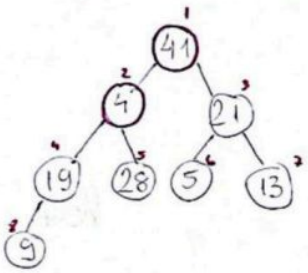
3) sink(3) // 5



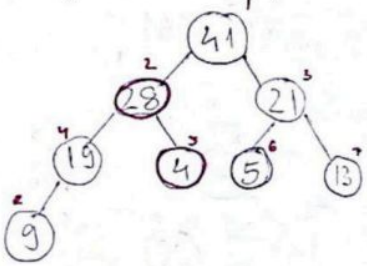
4) delMax(); // 49
exchange 49 and 4



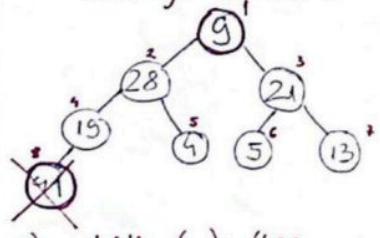
5) sink(1) // 4



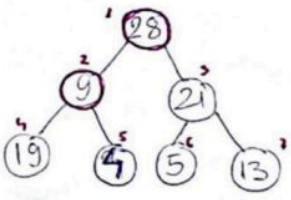
6) sink(2) // 4



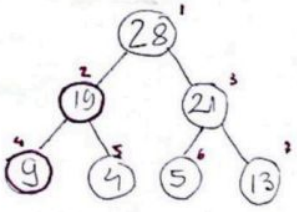
7) delMax(); // 41
exchange 41 and 9



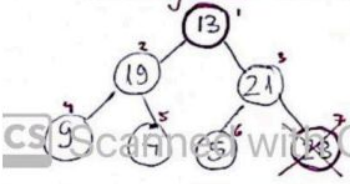
8) sink(1) // 9



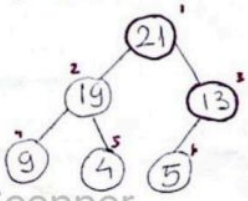
9) sink(2) // 9



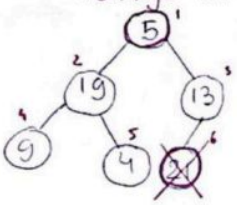
10) delMax(); // 28
exchange 28 and 13



11) sink(1) // 13

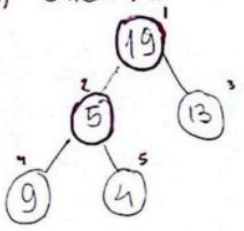


12) delMax(); // 21
exchange 21 and

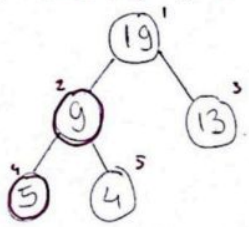


Scanned with CamScanner

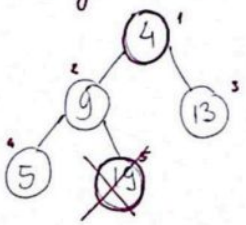
13) sink(1) // 5



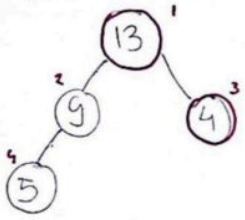
14) sink(2) // 5



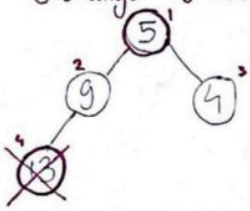
15) delMax(); // 19
exchange 19 and 4



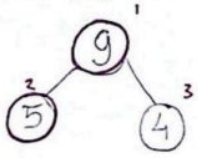
16) sink(1) // 4



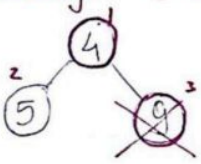
17) delMax(); // 13
exchange 13 and 5



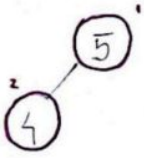
18) sink(1) // 5



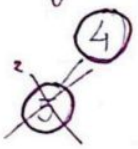
19) delMax(); // 9
exchange 9 and 4



20) sink(1) // 4



21) delMax(); // 5
exchange 5 and 4



array in sorted order

4	5	9	13	19	21	28	41	49	68

Link për visualizim të HeapSort: https://algostructure.com/sorting/heapsort.php